Advanced AC Circuit

Laboratory Exercise No: 2

Diode Rectifier Circuits

Introduction

The main objective of this laboratory exercise is to (1) consolidate the principle of the unidirectional conductivity of diodes and master the role of diodes in rectifier circuits. In this lab, you will learn to understand and grasp the composition, working principles, and waveform characteristics of three typical rectifier circuits: half-wave rectification, full-wave rectification (center-tapped), and bridge rectification. (2) explore the influence of filter capacitors on the output waveform of rectifier circuits and understand the basic design idea of "rectification + filtering" in power supply circuits. The core of a rectifier circuit is to use the "unidirectional conductivity" of diodes to convert the alternating current (AC) voltage that alternates between positive and negative into a unidirectional pulsating direct current (DC) voltage. In this experiment, a low-voltage AC power is obtained from the 220V 50Hz AC mains supply via a transformer and its output voltage will be used as the input AC power for the diode rectifier circuits. The principles of the three typical rectifier circuits are explained as follows:

Half-Wave Rectifier Circuit

It consists of one diode and a load resistor. When the AC input voltage is in the positive half-cycle, the diode is forward-biased and conducts, and current flows through the load to form the output; in the negative half-cycle, the diode is reverse-biased and cuts off, and there is no current through the load, so the output is 0. Therefore, the output waveform only retains the positive half-cycle (or negative half-cycle, depending on the direction of the diode) of the AC signal, with large pulsation and low efficiency (only 40.6%).

Full-Wave Rectifier Circuit (Center-Tapped)

It consists of two diodes, a center-tapped transformer, and a load resistor. The voltages at both ends of the secondary coil of the transformer are equal in magnitude and opposite in direction (with the center tap as the reference): in the positive half-cycle, one of the diodes conducts, and current flows from the center tap to one end of the secondary coil through the load; in the negative half-cycle, the other diode conducts, and current flows from the center tap to the other end of the secondary coil through the load. The output waveform retains both the positive and negative half-

cycles of the AC signal (both converted into unidirectional current), with a pulsation frequency twice that of half-wave rectification and higher efficiency (approximately 81.2%).

Bridge Rectifier Circuit

It consists of four diodes (or an integrated rectifier bridge) and a load resistor (no need for a center-tapped transformer). In the positive half-cycle, two diagonal diodes conduct, and current forms a path through the load; in the negative half-cycle, the other two diagonal diodes conduct, and the direction of current through the load remains unchanged. The output waveform is consistent with that of full-wave rectification (bidirectional pulsation is converted into unidirectional), but it does not require a special transformer, has a more flexible structure, and is the most commonly used rectification method in practical circuits.

Role of Filter Capacitor

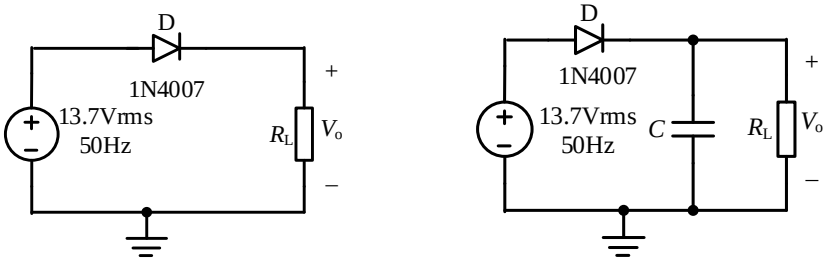
The output after rectification is "unidirectional pulsating DC" (containing a large amount of AC ripple). When an electrolytic capacitor is connected in parallel across the load, the capacitor will charge at the peak of the voltage and discharge at the valley, supplementing the load current, making the output waveform smoother (reducing ripple), and the effective value of output voltage is closer to the peak value of the input AC signal.

1. Pre-Lab Work

Use Multisim to simulate the rectifier circuits in Figure 1 to Figure 2.

(1) Record the waveforms from the outputs when R_L=1kΩ, 10kΩ, C=100μF, 1000μF respectively.
(2) Measure the ripple range of the output voltage if there is any.
(3) Measure the DC voltages from the outputs.
(4) Analyze the results, what conclusions can you get?

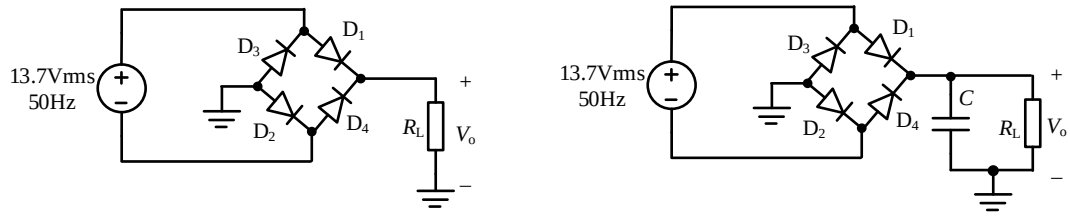
(a) Half wave rectification



(a) (b)

Figure 1. Half wave rectifiers

(b) Bridge rectification



(a) (b)

Figure 2. Bridge rectifiers

Table 1 Simulation measurement

Circuit Type	Filter Capacitor	Input AC Voltage (V AC, Effective Value)	Output Average Voltage (V DC) R_L=1kΩ/ 10kΩ		Output Waveform Characteristics (Hand-drawn/Description)	Ripple Condition (Large/Medium/Small)
Half-wave rectification	None	13.7				
Half-wave rectification	100μF	13.7				
Half-wave rectification	1000μF	13.7				
Bridge rectification	None	13.7				
Bridge rectification	100μF	13.7				
Bridge rectification	1000μF	13.7				

2. Lab Section 1

2.1 Verification of Diode Unidirectional Conductivity (Basic Preparation)

Use the "diode range" of the multimeter to measure the forward and reverse voltage drops of the 1N4007: connect the red test lead from the multimeter to the anode (long lead) of the diode and the black test lead to the cathode (short lead), and record the forward voltage drop V_F (approximately 0.6-0.7V); reverse the test leads and record the reverse voltage drop (display "overflow" or close to infinity). Get your conclusion.

2.2 Measurement of the AC power

(a) Use the multimeter to measure the output voltage V_{rms} and the frequency f from the given transformer. Calculate the peak voltage V_p.

(b) Use CH1 channel of the oscilloscope to show the wave form of the AC power (the output from the transformer). Notice that the wave form might not exactly the same as that normally shown from the Function Generator.

2.3 Construction and Testing of Half-Wave Rectifier Circuit

Circuit Construction: Connect the circuit as shown in Figure 1. Test the circuit (a) and then circuit (b).

(a) Test the circuit (a)

Use the "CH1 channel" of the oscilloscope to measure the input AC power waveform and record the waveform V_p and the frequency f.

Use the "CH2 channel" of the oscilloscope to measure the output waveform across the load, observe and record the waveform.

Use the "AC voltage range" of the multimeter to measure the voltage across the load (output average voltage V_o half-wave), and record the data (theoretical value: $V_{o\ \text{half-wave}} \approx 0.45 \times V_s$, where V_s is the effective value of the input AC power, i.e., approximately 0.45×13.7=6.17V. Considering the voltage drop across the diode, the output voltage is around $V_{o\ \text{half-wave}} \approx 0.45 \times V_s - V_F$.

(b)Testing with filter capacitor: Connect a 100μF electrolytic capacitor in parallel across the load (pay attention to the positive and negative poles: the positive pole of the capacitor is connected to the cathode side of the diode, and the negative pole is connected to the ground side of the load).

Use the oscilloscope to observe the waveform across the load and record the change.

Use the multimeter (DC voltage range) to measure the output voltage and record the data (the voltage at this time is close to the peak value of the input AC, approximately $1.414V_{rms} - V_F$).

2.3 Construction and Testing of Bridge Rectifier Circuit

Connect the circuit shown in Figure 3 when R_L=1kΩ and then 10kΩ.

(a)Testing circuit (a):

Use the oscilloscope to measure the output waveform and record the waveform.

Use the multimeter to measure the output average voltage (theoretical value is $V_o \approx 0.9 \times V_s - 2V_F$) and record the data.

(b) Testing circuit (b): Connect 100μF (and then 1000μF) electrolytic capacitor in parallel with the load, observe the waveform, measure the output voltage under different capacitors, and record the data.

Table 2 Practical measurement

Circuit Type	Filter Capacitor	Input AC Voltage (V AC, Effective Value)	Output Voltage (V DC) R_L=1kΩ/ 10kΩ		Output Waveform Characteristics (Hand-drawn/Description)	Ripple Condition (Large/Medium/Small)
Half-wave rectification	None	13.7				
Half-wave rectification	100μF	13.7				
Half-wave rectification	1000μF	13.7				
Bridge rectification	None	13.7				
Bridge rectification	100μF	13.7				
Bridge rectification	1000μF	13.7				

3. Thinking Questions (Deepen Understanding)

(1) In a half-wave rectifier circuit, if the positive and negative poles of the diode are reversed, what will happen to the output waveform? If the diode is damaged (short-circuited or open-circuited), what phenomenon will occur in the circuit?

(2) Is the larger the capacity of the filter capacitor, the better the filtering effect? Why can't the capacitor be infinitely large in practical circuits? (Hint: Capacitor charging current, cost, volume)

(3) If the input AC voltage in the experiment is 220V AC (mains supply), what parameters should be considered when selecting a rectifier diode? (Hint: Reverse withstand voltage, forward current)

4. Lab Report Submission

The report can be completed in pairs.